



iPLUG
User Guide
Network Resilience Plugins

Document information

Task	1.4
Document Name	User Guide Network Resilience Plugins
Version	01
Release date	30/06/2024
Dissemination level	Public
Status	Submitted
Authors	Montserrat Montalà-Palau, Marc Cheah Mañe and Oriol Gomis-Bellmunt (UPC)



This project has received funding from the European Union's Horizon Europe Research and Innovation programme under **Grant Agreement No 101069770**.

Document history:

Version	Date of issue	Content and changes	Edited by
01	04/06/2024	User Guide	UPC

Table of contents

1	UserGuide	6
1.1	Installation	7
1.1.1	Permission access	7
1.1.2	General Requirements	7
1.1.3	Recommended additional Plugins	7
1.1.4	Specific requirement for → Impact Evaluation	7
1.1.5	Plugin Installation steps	7
1.2	Usage	7
1.3	Hazards Combination	8
1.4	Impact Evaluation	10
1.5	Risk Assessment and Resilience Index	13

List of Figures

1	Dialog Box of CITCEA Probability Analysis Plugin	9
2	Demand Profiles hourly adjusting factor	10
3	Dialog Box of CITCEA Probability Analysis Plugin	11
4	Dialog Box of CITCEA Resilience Index Plugin	13

List of Tables

1	Standard Lines. Additionally, the standard lines from pandapower can be specified by the user: pandapower_standard_lines	11
---	--	----

1 UserGuide

Three QGIS Plugin are developed in the context of the iPlug project to address the network's resilience.

1. **Hazard Combination** → This Plugin creates a new layer providing the probability of occurrence of a combined event from simple events. The user should indicate the type of relation between events: dependent or independent.
2. **Impact Evaluation** → This Plugin determines the impact of each element based on an OPF analysis. The user should introduce one layer for buses and one for power lines. The Attribute Table of each layer should include the required parameters to perform the OPF analysis. Besides, the user can select the type of OPF (AC or DC) and the simulation (24-hour profile or instantaneous). As a result, the Plugin created two new layers, one for buses and one for power lines. Each new layer includes, in the attribute table, the impact of each element in power or energy, depending on the simulation type selected. Finally, the Plugin provides a spreadsheet with the same information.
3. **Risk Assessment and Resilience Index** → This Plugin evaluates power/energy at risk in each element based on the hazards that might affect a system and its vulnerabilities. First, the user should indicate the year's season to perform the analysis. Then, the maximum power line's segment length. Finally, the user should upload a spreadsheet with the system's vulnerabilities.

Please, follow the stages described in this User Guide when using the Plugins, otherwise, the results may not be as expected.

1.1 Installation

1.1.1 Permission access

The Plugins are available in a GitHub private repository. We ask new users to request permission access from the iPlug administration to continue.

You can obtain permission by emailing montserrat.montala@upc.edu indicating an Affiliation Link and the purpose of the request.

1.1.2 General Requirements

- **QGIS $\geq$ 3.0:** Download from [QGIS OfficialSite](#).

1.1.3 Recommended additional Plugins

- **Plugin Reloader:** Download from [Plugin Reloader OfficialSite](#).

1.1.4 Specific requirement for → Impact Evaluation

- **Pandapower $\geq$ 6.4:**
 - Open OSGeo4W shell (packed with QGIS in the start menu)
 - Install the library using "pip", following the instructions at the [Pandapower OfficialSite](#).

1.1.5 Plugin Installation steps

- Download from the repository the required Plugin.
- Unzip the folder.
- Save the folder in the QGIS' Active User Profile.
 - Linux:

C:\Users\USER\AppData\Roaming\QGIS\QGIS3\profiles\default\python\Plugins\ORStools
 - Windows:

C:\Users\USER\AppData\Roaming\QGIS\QGIS3\profiles\default\python\Plugins\ORStools
 - Mac OS:

Library/Application Support/QGIS/QGIS3/profiles/default/python/Plugins/ORStools

If not found, the QGIS' Active User Profile Folder can be accessed through "QGIS" ► "Settings" ► "User Profile" ► "Open Active Profile Folder"

1.2 Usage

- Before using a Plugin, using the Reload Plugin through "Plugin Reloader" is recommended.
- Follow the instructions detailed in Sections 1.3 to 1.5 for each Plugin.

1.3 Hazards Combination

A resilience analysis should consider all potential hazards that might affect the system. Different institutions and organizations provide geolocated information about simple hazards. The user should store all simple hazards in a **QgsTreeLayer** called *Hazards*, otherwise the Plugin will not consider them. The Layers included in the QgsTreeLayer should contain at least the following information in its Attribute Table:

1. Field *SUMMER* → probability of occurrence during summer.
2. Field *WINTER* → probability of occurrence during winter.
3. Field *AUTUMN* → probability of occurrence during autumn.
4. Field *SPRING* → probability of occurrence during spring.

Simple hazards can simultaneously occur, increasing the vulnerabilities of the system. In that context, the Hazard Combination Plugin results in a new layer that includes the probability of having the combined events in each season of the year.

Figure 1 shows the Dialog Box of this Plugin. First, the user should select the type of combination between events: dependent or independent. Second, the user should choose *Event A* and *Event B* among all layers included in the QgsTreeLayer called *Hazards*.

The Plugin will unlock the $P(A|B)$ boxes when the user chooses dependent events. The user should introduce how the probability of having *Event A* varies once *Event B* occurs. Introduce this variation using per unit (p.u.) notation: 1 signifies no variation, 0.5 represents half the probability, and 2 indicates double the probability.

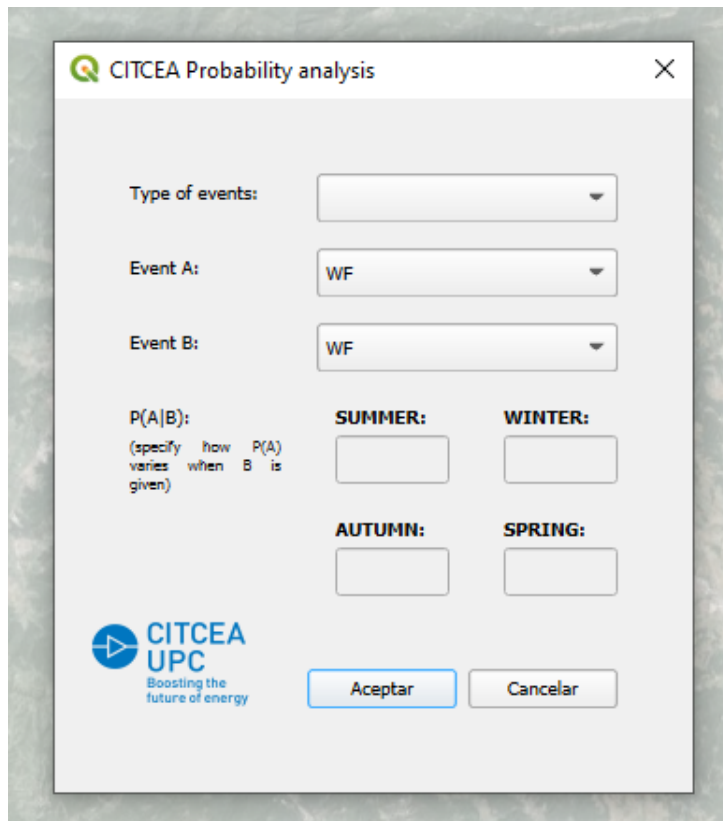


Figure 1: Dialog Box of CITCEA Probability Analysis Plugin

The Plugin adds a new layer to the hazard database, including the combined probability. The layer should be moved to the QgsTreeLayer *Hazards* after being validated. Once included, the user can select the layer in subsequent event combinations.

1.4 Impact Evaluation

An OPF analysis of the system can provide the impact of each element on the network. The Impact Evaluation Plugin generates a new layer that includes the element's impact.

A QgsTreeLayer called *System_TBA* should include the layers of the network. At least two layers are requested: one for buses and one for power lines. The Layers included in the QgsTreeLayer should contain at least the following information in the Attribute Table:

1. Buses Layer:

- (a) Field *Voltage_kV* → voltage level in kV.
- (b) Field *P_GEN* → power injected from a generator to the bus in MW.
- (c) Field *Type_GEN* → the generators can be modeled as a PQ or PV. A PQ generator provides constant active and reactive power while a PV provides constant active power while fixing the voltage in a given setpoint.
- (d) Field *P_LOAD* → power demanded to the bus in MW.
- (e) Field *GRID* → 1 in case of being connected to the external grid, 0 otherwise.
- (f) Field *DEM_PROF* → this item is only necessary if an hourly OPF is expected. The user should indicate the type of daily profile from: *COMMERCIAL*, *RESIDENTIAL* or *INDUSTRIAL*. The power specified in *P_GEN* is assumed as the maximum power and scaled according to Figure 2.

2. Power Lines Layer:

- (a) Field *STD_Line* → Type of standard type among the types included in Table 1
- (b) Field *PARALLEL* → number of parallel line systems

The *From* and *To* buses, as well as the *Length*, are automatically determined through the geolocated data.

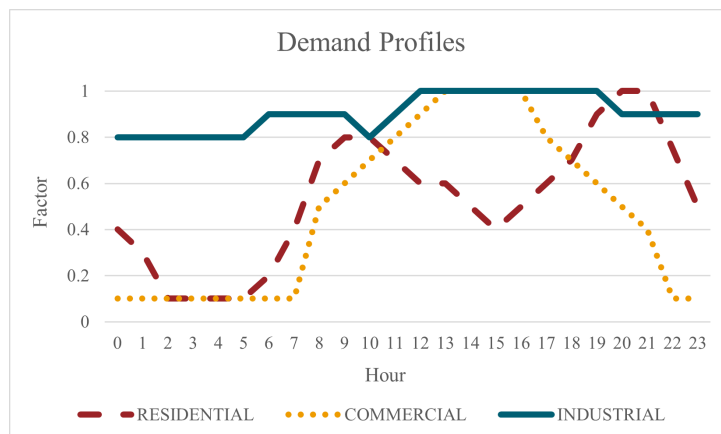


Figure 2: Demand Profiles hourly adjusting factor

STD.Line	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max.i_ka
ES.01	0.0460	0.1000	0	0.630
ES.02	0.1610	0.1050	0	0.415
ES.03	0.2620	0.1210	0	0.315
ES.04	0.2620	0.1160	0	0.315
ES.05	0.2620	0.1170	0	0.300
ES.06	0.2700	0.1260	0	0.305
ES.07	0.4030	0.1200	0	0.250
ES.08	0.6140	0.3740	0	0.200
ES.09	0.7900	0.3800	0	0.180
ES.10	1.0736	0.3850	0	0.140
ES.11	1.2150	0.4000	0	0.130

Table 1: Standard Lines. Additionally, the standard lines from pandapower can be specified by the user: `pandapower_standard_lines`

The Dialog Box of the Plugin is represented as Figure 3.

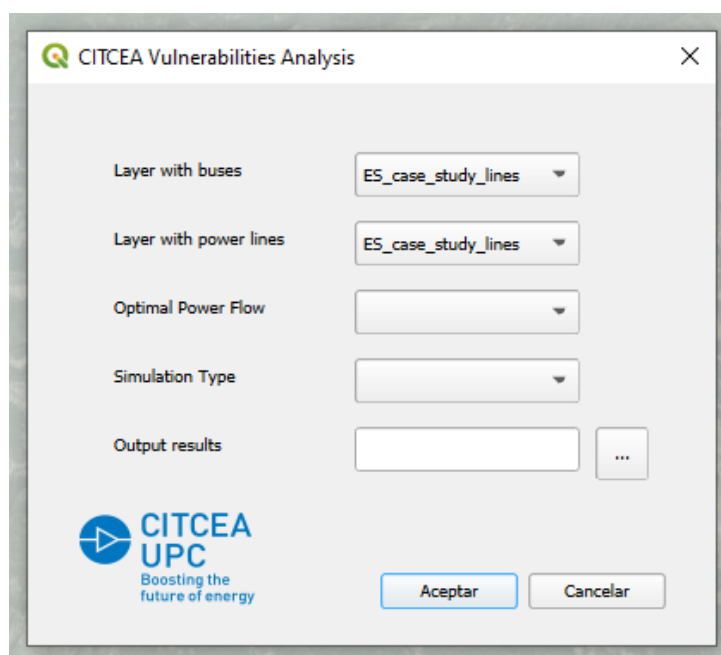


Figure 3: Dialog Box of CITCEA Probability Analysis Plugin

First, the user should select the layers with buses and power lines in the first and second dropdown lists, respectively. The list shows all layers included in the `QgsTreeLayer System.TBA`. Second, the user can select between a DC and AC OPF depending on the system characteristics and the expected accuracy. Third, the user should choose between instantaneous or hourly OPF. Finally, the user should name a .csv file to save the element's impact obtained.

As a result, the Plugin generates two new layers, one for buses and one for power lines. The layers contain the same data in the Attribute Table as the input layers and a new Field called *Impact*. The Field *Impact* includes the amount of energy or power that would be unsupplied in case of losing the element.

Impact is quantified in terms of power if the user selects an instantaneous analysis. Additionally, the .csv includes a list of elements and their impact.

1.5 Risk Assessment and Resilience Index

A resilience analysis should consider all hazards affecting a system and its vulnerabilities. The Resilience Analysis Plugin enables risk assessment and the calculation of the system's resilience index. Figure 4 shows the Plugin Dialog Box.

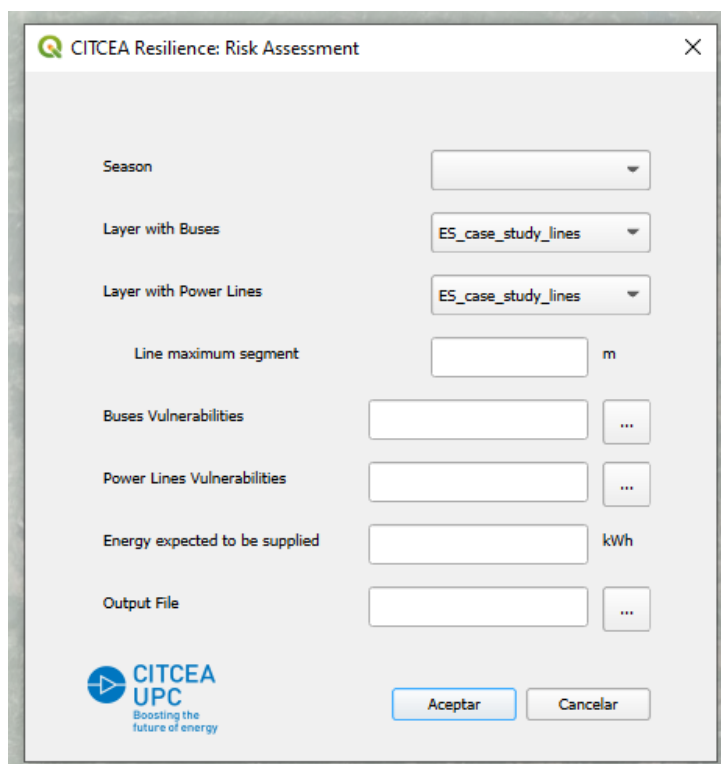


Figure 4: Dialog Box of CITCEA Resilience Index Plugin

First, the user should select the year's season to analyze. Second, the user should choose the layer with buses and power lines in the dropdown lists. The dropdown lists include all layers saved in the `QgsTreeLayer System.TBA`. Third, the user can specify the maximum length of each segment of the power lines. In case of not expecting a segmentation in power lines, introduce a value a high value. Fourth, the user should introduce a .csv file with the vulnerabilities of the buses and power lines, respectively. Fifth, the user should manually introduce the sum of total impacts. Finally, the user should specify the .csv file to save the risk assessment results.

Regarding the .csv of the vulnerabilities, the minimum content should be:

1. Column name: *Name* → The name of each bus / power line. The elements should have the same name used in the layers included in `QgsTreeLayer System.TBA`. The name of each element should be repeated for each vulnerability, i.e. if an element has 10 vulnerabilities, the .csv should include 10 rows and each row should include the element's name.
2. Column name: *VULNERABILITY* → vulnerability's name. The user should specify each vulnerability in a different row.

3. Column name: *hazard-name_likelihood* → As many columns as hazards included in the QgsTreeLayer *Hazards* need to be included replacing *hazard-name* by the layer's name. In each row, the user should specify the likelihood of the vulnerability affecting the system once the hazard occurs.
4. Column name: *hazard-name_damage* → As many columns as hazards included in the QgsTreeLayer *Hazards* need to be included replacing *hazard-name* by the layer's name. In each row, the user should specify the impact that the element has on the system. Those impacts can be estimated through the Plugin detailed in Section 1.4.

This Plugin adds two more layers to the QGIS project for buses and power lines. The Attribute Table of each layer includes the name of the element and the total power or energy risked in it. In the case of power lines, each segment is an independent feature in the resulting layer.

Finally, the .csv file includes more detailed information. The user can check the amount of power or energy risked in each element because of each vulnerability. This information might be helpful to assess the introduction of mitigation action in the system.



This project has received funding from the European Union's Horizon Europe research and innovation programme under grant agreement No. 101069770